# The Difficulty of Passive Learning in Deep Reinforcement Learning

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# Introduction

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Setting: Offline reinforcement learning, learning to act from observational data without active environmental interaction.

Challenge: extrapolation error, the value of missing or under-represented state-action pairs in the dataset can be over-estimated/under-estimated, it cannot be self-corrected via exploitation/exploration during interaction with the environment.

Experimental paradigm: tandem learning, empirical analysis of the difficulties in offline reinforcement learning.

# Motivation

- Psychology (Held and Hein's [1963] experiment): coupling two young animal subjects' movements and visual perceptions to ensure that both receive the same stream of visual inputs, while only one can actively shape that stream by directing the pair's movements.

- Tandem RL setup: pairing an 'active' and a 'passive' agent in a training loop where only the active agent drives data generation, while both perform identical learning updates from the generated data.

- Decoupling learning dynamics from its impact on data generation, while preserving the non-stationarity of the online learning setting.

# Experimental Setting

- The ‘tandem’ analogy is that of two riders, both of whom experience the same route, while only the front rider gets to decide on direction.

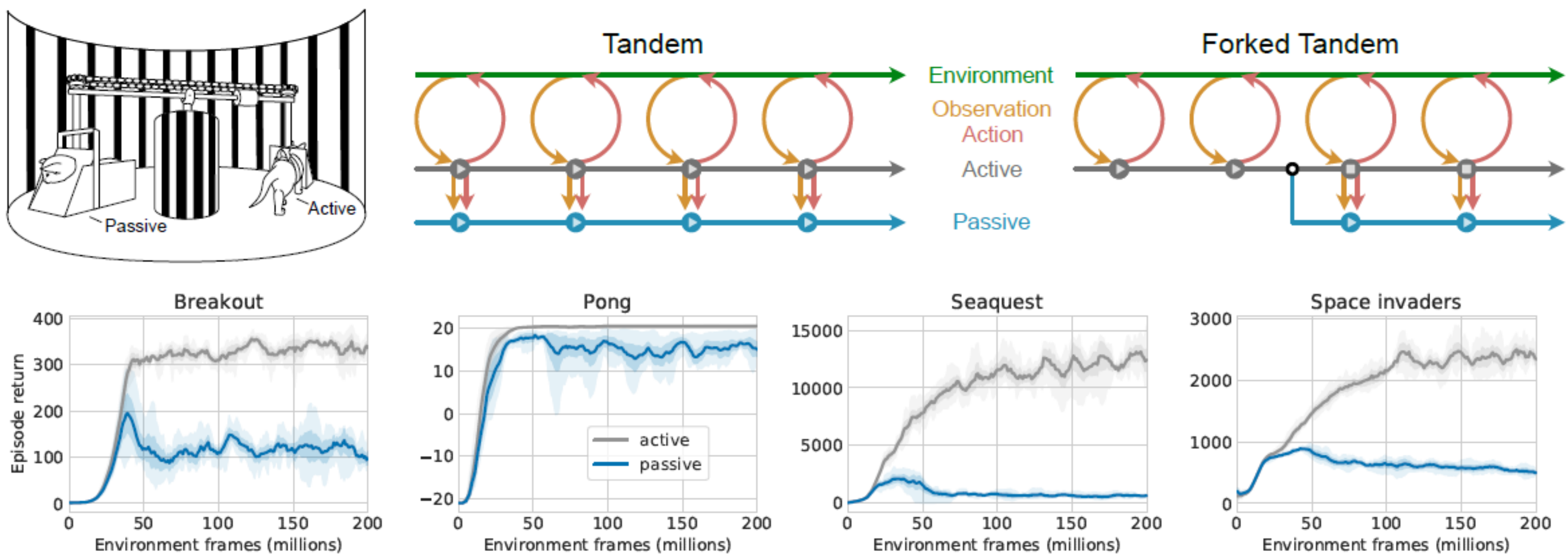


Figure 1: (**top-left**) Held and Hein [1963] experiment setup. (**top-right**) Illustrations of the Tandem and Forked Tandem experiment setups. (**bottom**) Tandem (active and passive) performance on 4 selected Atari domains. In all figures, *active* agent performance is shown in **gray**.

# Experimental Setting

Tandem effect:

- The passive learner fails to learn from the very data stream that is demonstrably sufficient for its architecturally identical active counterpart.

Tandem RL setting:

- Tandem: Active and passive agents start with independently initialized networks, and train on an identical sequence of training batches in the exact same order.
- Forked Tandem: start out with identical network weights, the active agent receives no further training but continues to generate data.

# Experimental Setting

Three potential contributing factors:

- Bootstrapping (B): bootstrapping from poorly estimated values.
  (1) Target Q  (2) Target policy

- Data Distribution (D): insufficient coverage of sub-optimal actions.
  (1) The exploration parameter $\epsilon$ (2) Sticky actions (3) Replay size
  (4) Fixed policy 5) Fixed replay (6) On-policy evaluation
  (7) Self-generated data

- Function Approximation (F): wrongly extrapolate the values of underrepresented state-action pairs.
  (1) Optimization (2) Function class

Algorithm and environment：
- Double-DQN
- Breakout, Pong, Seaquest, Space invaders

# Experiment and Analysis

The Role of Bootstrapping

- Double-DQN: $Q(s,a) \leftarrow r + \gamma \bar{Q}(s', \mathrm{argmax}_{a'} Q(s', a'))$

$$Q_P(s,a) \leftarrow \begin{cases} r + \gamma \bar{Q}_P(s', \arg\max_{a'} Q_P(s', a')) & \text{Vanilla Tandem DQN} \\ r + \gamma \bar{Q}_A(s', \arg\max_{a'} Q_P(s', a')) & \text{Same Target } Q \\ r + \gamma \bar{Q}_P(s', \arg\max_{a'} Q_A(s', a')) & \text{Same Target } \pi \\ r + \gamma \bar{Q}_A(s', \arg\max_{a'} Q_A(s', a')) & \text{Same Target } \pi \& Q \end{cases}$$

# Experiment and Analysis

The Role of Bootstrapping

- The use of the active value functions as targets reduces the active-passive gap by only a small amount.

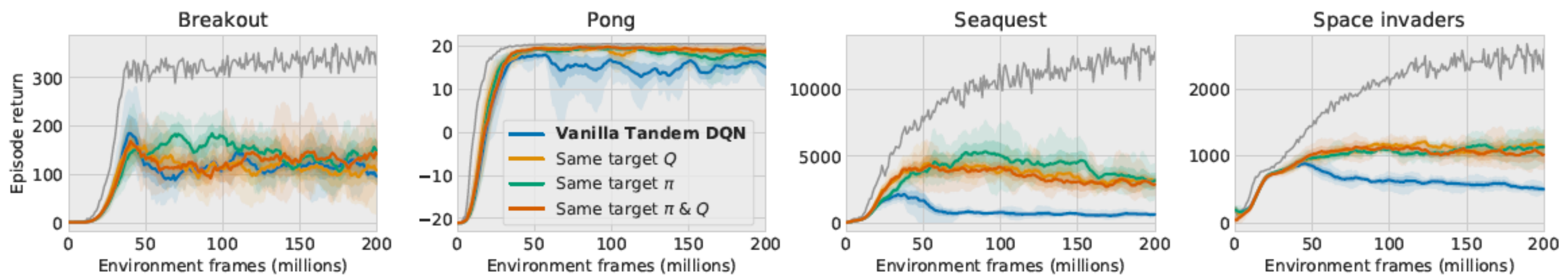


Figure 2: Active vs. passive performance when using the active agent's target policy and/or value function for constructing passive bootstrapping targets.

# Experiment and Analysis

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## The Role of Bootstrapping

- Passive networks trained using the active network's bootstrap targets do not over-estimate compared to the active network at all.
- Value over-estimation itself is not the fundamental cause of the tandem effect, and that (B) plays an amplifying, rather than causal role.

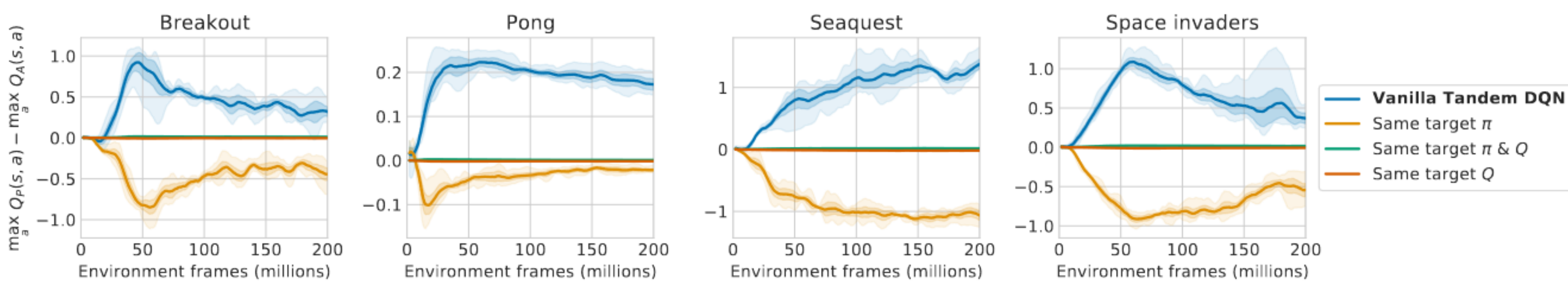


Figure 16: Q-value over-estimation by the passive network compared to the active one in Tandem DQN with varying bootstrap targets. It can be seen that the passive network in the vanilla Tandem DQN setting tends to over-estimate values (compared to the active one), which is almost perfectly mitigated by using the same bootstrap target *values* as the active network, and in fact reversed when using the same target *policy* (but not the same target values). Note that in all four configurations the passive agents substantially under-perform their active counterparts (Fig. 2), showing that bootstrapping-amplified over-estimation is only part, not the main cause of the tandem effect.

# Experiment and Analysis

The Role of the Data Distribution

- The exploration parameter $\epsilon$: the performance improves when the active behavior policy's stochasticity increases (coverage of non-argmax actions).
- Sticky actions: no substantial impact on the tandem effect (environment stochasticity).

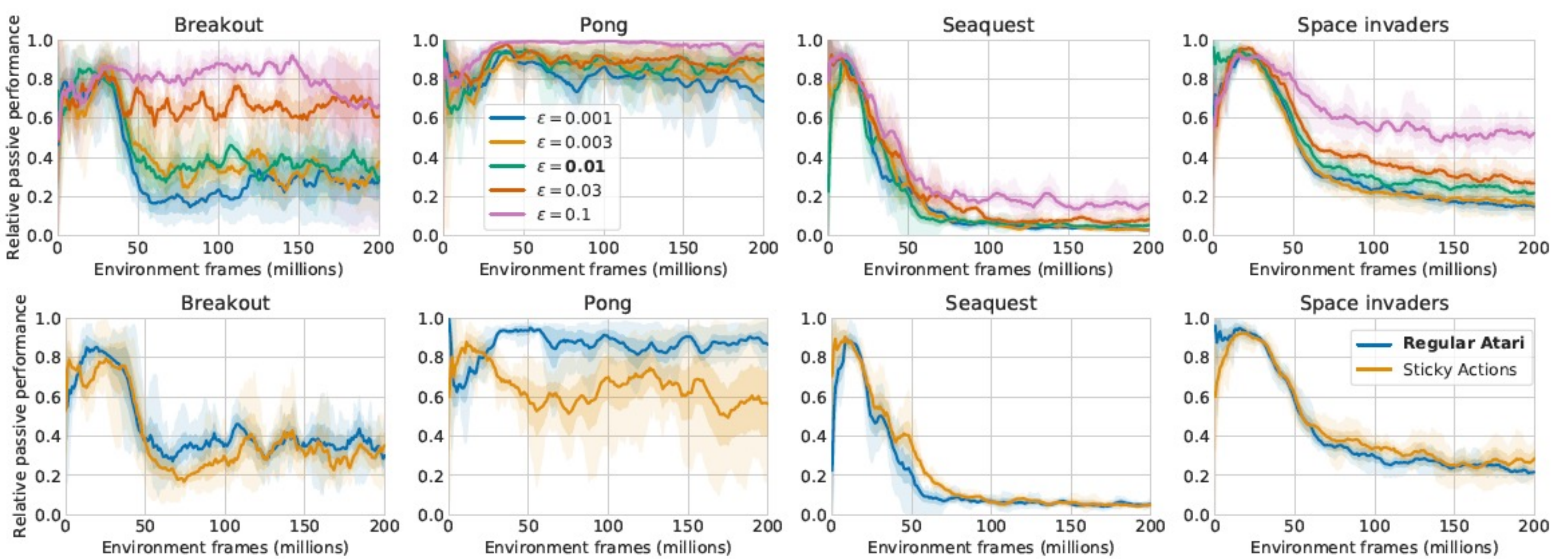


Figure 3: Passive as fraction of active performance for varying active $\varepsilon$-greedy behavior policies (**top**); regular Atari vs sticky-actions Atari (**bottom**). We report *relative* passive performance, as active performance varies across configurations. See Appendix Figs. 17 & 18 for absolute performance.

# Experiment and Analysis

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## The Role of the Data Distribution

- Replay size: larger replay buffer somewhat mitigates the passive agent's under-performance, mostly slows down rather than prevents from eventually under-performing.

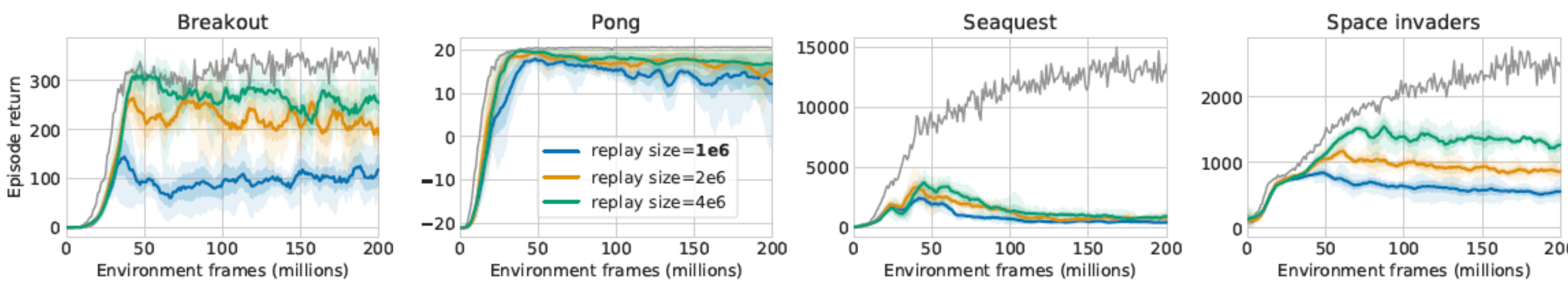


Figure 4: Active vs. passive performance for different replay sizes (for passive agent).

# Experiment and Analysis

The Role of the Data Distribution

- Fixed policy (Forked Tandem): the tandem effect is even exacerbated. Instability appears to be inherent in the learning process itself.
- Fixed replay (samples from a number of different policies): a larger replay buffer, containing samples from more diverse policies, provide an improved coverage of non-greedy actions, reducing the tandem effect.

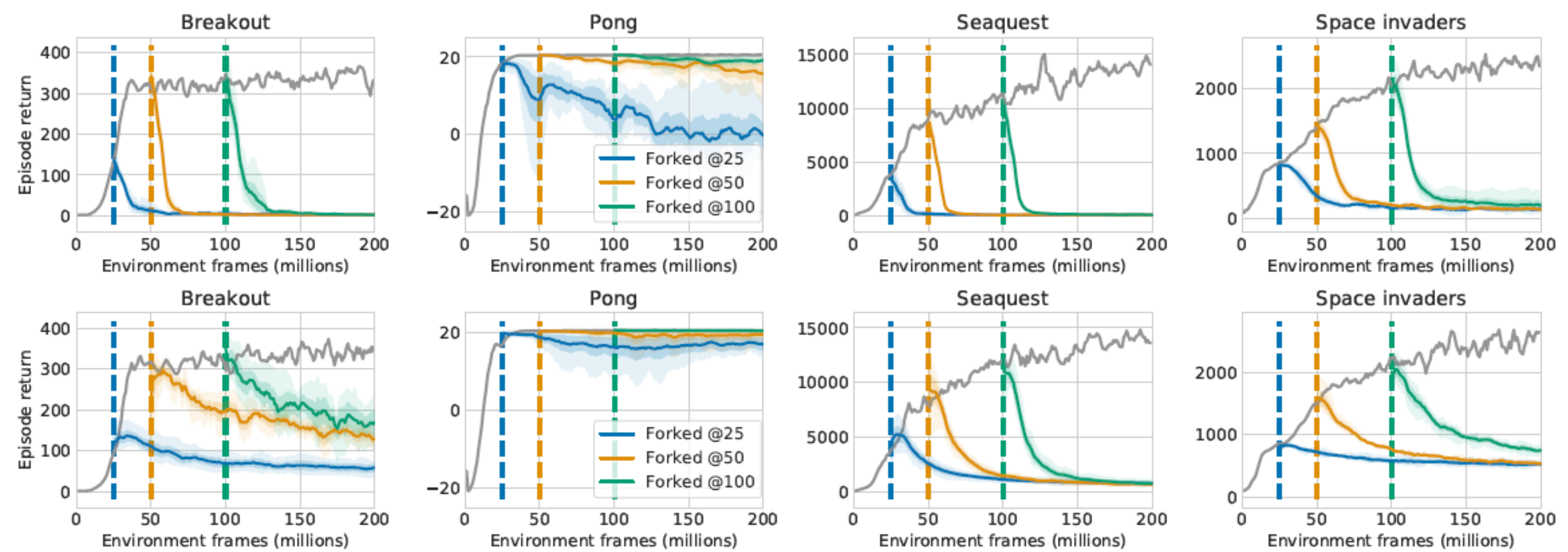


Figure 5: Performance of a Forked Tandem DQN, training passively after freezing its data generating policy (**top**) or its replay buffer (**bottom**). Vertical lines indicate forking time points.

# Experiment and Analysis

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The Role of the Data Distribution

- On-policy evaluation

Whether off-policy Q-learning itself is to blame for the collapse of passive performance?

- SARSA
- Supervised regression on the Monte-Carlo returns

Q(a_1,a_2|s)

(s,a_1,r,s')

Q(a_1|s) = r + $\gamma$Q(a'|s')

argmax_a Q(s)

**Q-learning (off-policy TD control) for estimating $\pi \approx \pi_*$**

Algorithm parameters: step size $\alpha \in (0, 1]$, small $\varepsilon > 0$
Initialize $Q(s, a)$, for all $s \in \mathcal{S}^+, a \in \mathcal{A}(s)$, arbitrarily except that $Q(terminal, \cdot) = 0$

Loop for each episode:
  Initialize $S$
  Loop for each step of episode:
    Choose $A$ from $S$ using policy derived from $Q$ (e.g., $\varepsilon$-greedy)
    Take action $A$, observe $R$, $S'$
    $Q(S, A) \leftarrow Q(S, A) + \alpha\big[R + \gamma \max_a Q(S', a) - Q(S, A)\big]$
    $S \leftarrow S'$
  until $S$ is terminal

**Sarsa (on-policy TD control) for estimating $Q \approx q_*$**

Algorithm parameters: step size $\alpha \in (0, 1]$, small $\varepsilon > 0$
Initialize $Q(s, a)$, for all $s \in \mathcal{S}^+, a \in \mathcal{A}(s)$, arbitrarily except that $Q(terminal, \cdot) = 0$

Loop for each episode:
  Initialize $S$
  Choose $A$ from $S$ using policy derived from $Q$ (e.g., $\varepsilon$-greedy)
  Loop for each step of episode:
    Take action $A$, observe $R$, $S'$
    Choose $A'$ from $S'$ using policy derived from $Q$ (e.g., $\varepsilon$-greedy)
    $Q(S, A) \leftarrow Q(S, A) + \alpha\big[R + \gamma Q(S', A') - Q(S, A)\big]$
    $S \leftarrow S'$; $A \leftarrow A'$;
  until $S$ is terminal

# Experiment and Analysis

The Role of the Data Distribution

On-policy evaluation:

- Strengthening the roles of (D) and (F) while further weakening that of (B).
- Instability results from erroneous extrapolation by the function approximator, when the utilized data distribution provides inadequate coverage of relevant state-action pairs.

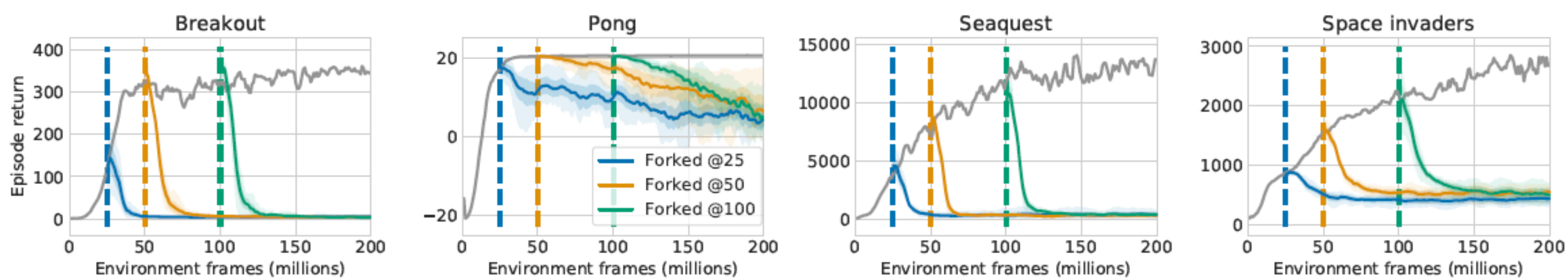


Figure 6: Passive performance in Forked Tandem DQN after policy evaluation with SARSA.

# Experiment and Analysis

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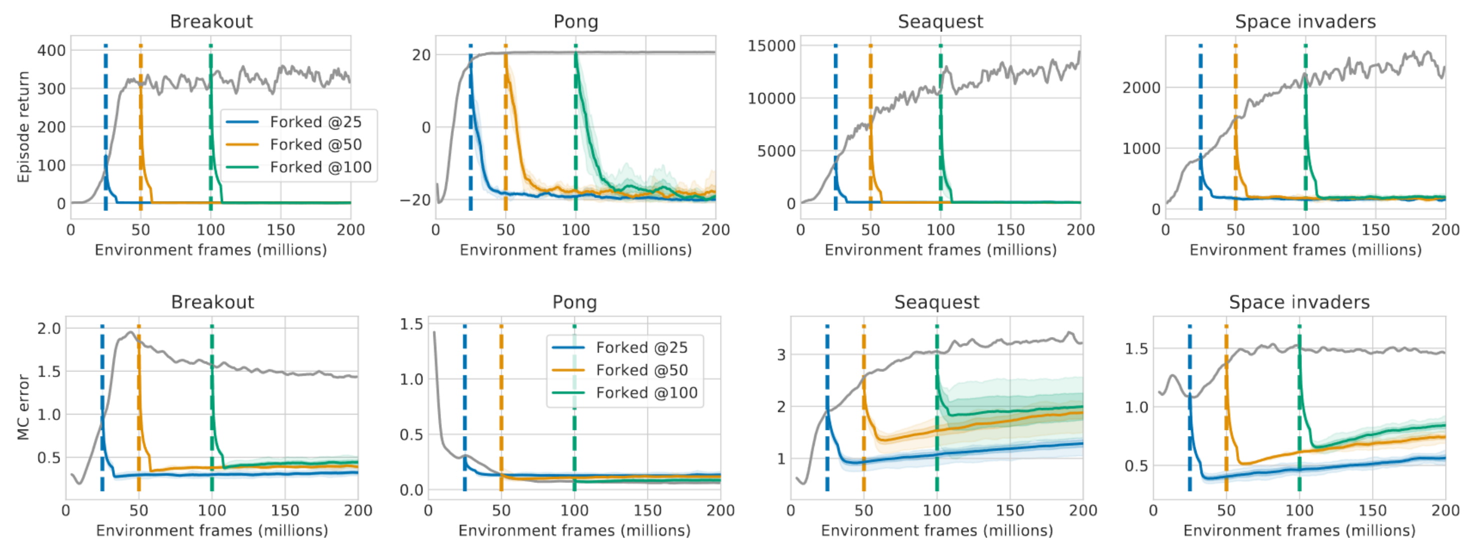


Figure 25: Forked Tandem DQN: Policy evaluation with Monte-Carlo return regression. (**top**) Active vs. passive control performance. (**bottom**) Average absolute Monte-Carlo error. The Monte-Carlo error is minimized effectively by the passive training, while the control performance of the resulting $\varepsilon$-greedy policy collapses completely.

# Experiment and Analysis

The Role of the Data Distribution

- Self-generated data (learn from mixed data): a moderate amount (10%-20%) of ‘own’ data yields a substantial reduction of the tandem effect, while a 50/50 mixture completely eliminates it.

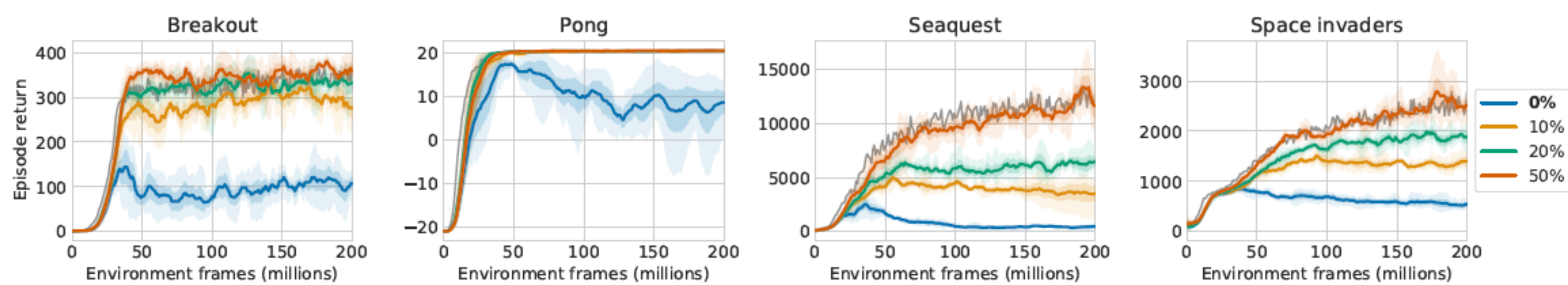


Figure 7: Passive performance for different amounts of self-generated data in the passive agent’s replay batches.

# Experiment and Analysis

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Role of Function Approximation

Optimization:

- Adam outperforms RMSProp, but the tandem effect itself is unaffected.
- More updates worsen the performance of the passive agent

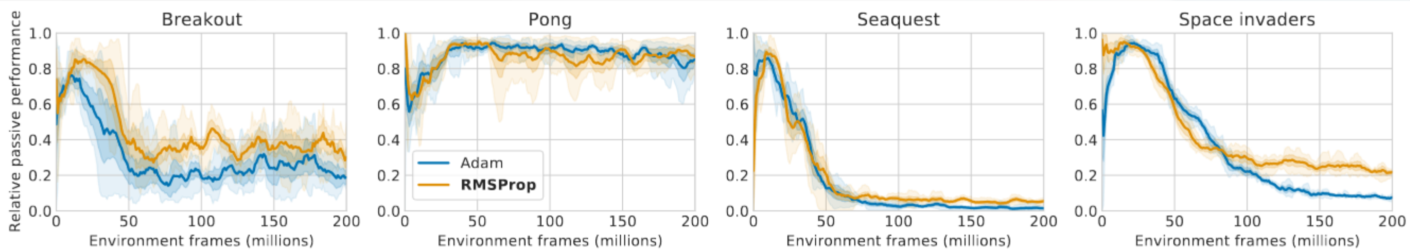


Figure 20: Tandem DQN with the Adam optimizer (instead of RMSProp) for both active and passive network optimization. (**top**) Adam: Active vs. passive performance. (**bottom**) Passive as fraction of active performance Adam vs. RMSProp. While the Adam optimizer improves both the active and passive performance of the Tandem DQN, the relative active-passive gap is not affected strongly.

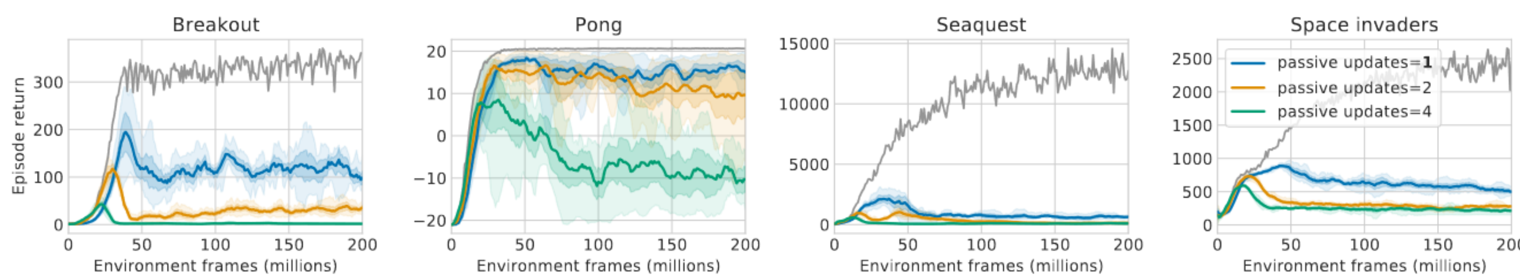


Figure 21: Active vs. passive performance for varying number of passive updates per active update.

# Experiment and Analysis

## Role of Function Approximation

Function class:

- The gap correlates positively with network depth, negatively with network width.
- Passive performance correlates with the number of tied layer.

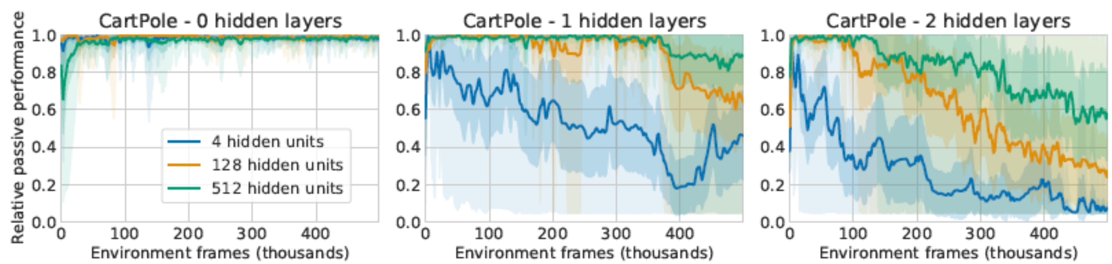


Figure 8: Passive performance as a fraction of active performance in CartPole: varying number of hidden layers and units.

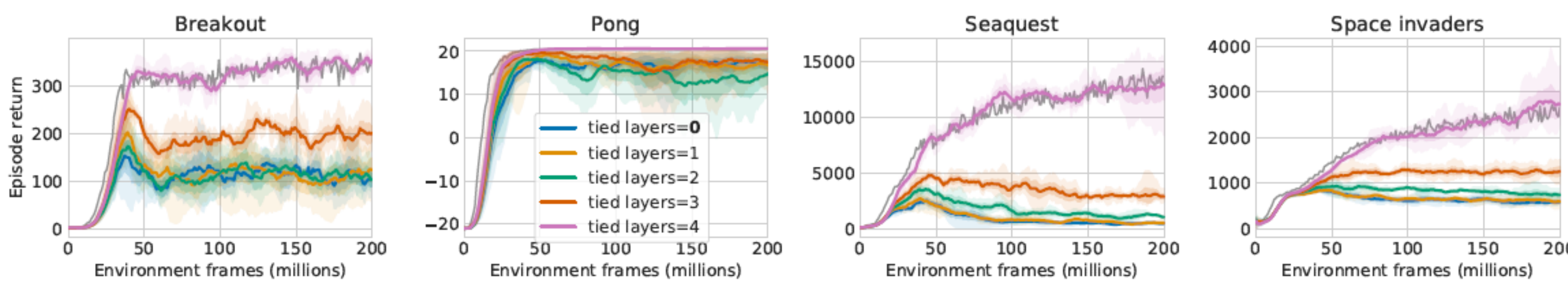


Figure 9: Active vs. passive performance, with first $k$ of 5 layers of active/passive networks shared.

# Conclusion

- Proposed tandem setting aims to decouple learning dynamics from its impact on behavior and the data distribution.

- Results indicate an empirically less critical role for bootstrapping, erroneous extrapolation or over-generalization by a function approximator trained on an inadequate data distribution as the crucial challenge.

- Highlights the importance of interactivity and ‘learning from your own mistakes’ in learning control.

# THANKS